OOAD LAB ASSESSMENT-1

Ticket Booking and Reservation System

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1. **Problem Description of the chosen system:**
   - Provide a comprehensive problem statement for the application.
   - Break down the description module-wise, explaining the scope, objectives, and challenges of each module

Problem Description of the Ticket Booking and Reservation System

The **Ticket Booking and Reservation System** is an online application designed to simplify the process of searching, reserving, booking, and managing tickets for various transportation services such as buses, trains, flights, or events. The system allows customers to search available schedules, check seat availability, reserve seats, complete secure payments, receive electronic tickets, and manage their bookings from a single platform. Administrators are responsible for maintaining schedules, routes, pricing, user information, and booking records.

The system consists of different users where **Customer** and **Admin** are specialized types of **User**. Every booking is associated with one customer and contains one or more tickets. Each ticket is linked to a schedule and an allocated seat. Payment is an essential part of booking confirmation and is processed through an external payment gateway. The system maintains booking history, ticket status, payment status, and refund information.

The application is designed to provide fast, secure, and user-friendly ticket booking while reducing manual effort, avoiding overbooking, and ensuring efficient management of schedules and reservations.

Module 1 – Login & User Management

Scope:

This module handles user authentication, registration, profile management, and role-based access to the system.

Objectives:

- Allow users to register and log in securely.
- Authenticate users based on their roles.
- Manage user profiles and passwords.
- Ensure secure access to system functionalities.

Challenges:

- Secure password storage and authentication.
- Prevent unauthorized access.
- Handle forgotten password requests.
- Manage multiple user roles (Customer and Admin).

Module 2 – Ticket Search & Reservation

Scope:

This module enables customers to search available tickets, view schedules, check seat availability, select seats, and reserve tickets before payment.

Objectives:

- Provide quick ticket search.
- Display accurate schedules and seat availability.
- Prevent double booking of seats.
- Create temporary reservations before payment.

Challenges:

- Maintaining real-time seat availability.
- Handling simultaneous reservations.
- Preventing duplicate bookings.
- Managing reservation expiry if payment is not completed.

Module 3 – Payment & Booking Management

Scope:

This module manages payment processing, booking confirmation, ticket generation, cancellation, and refunds.

Objectives:

- Secure payment processing.
- Generate electronic tickets.
- Maintain booking history.
- Process cancellations and refunds efficiently.

Challenges:

- Handling payment failures.
- Integrating with external payment gateways.
- Updating booking status correctly.
- Managing refund requests.

Module 4 – Admin & Schedule Management

Scope:

This module allows administrators to manage schedules, routes, pricing, users, seat inventory, bookings, and reports.

Objectives:

- Maintain accurate schedules.
- Manage ticket pricing.
- Monitor bookings.
- Generate reports for system analysis.

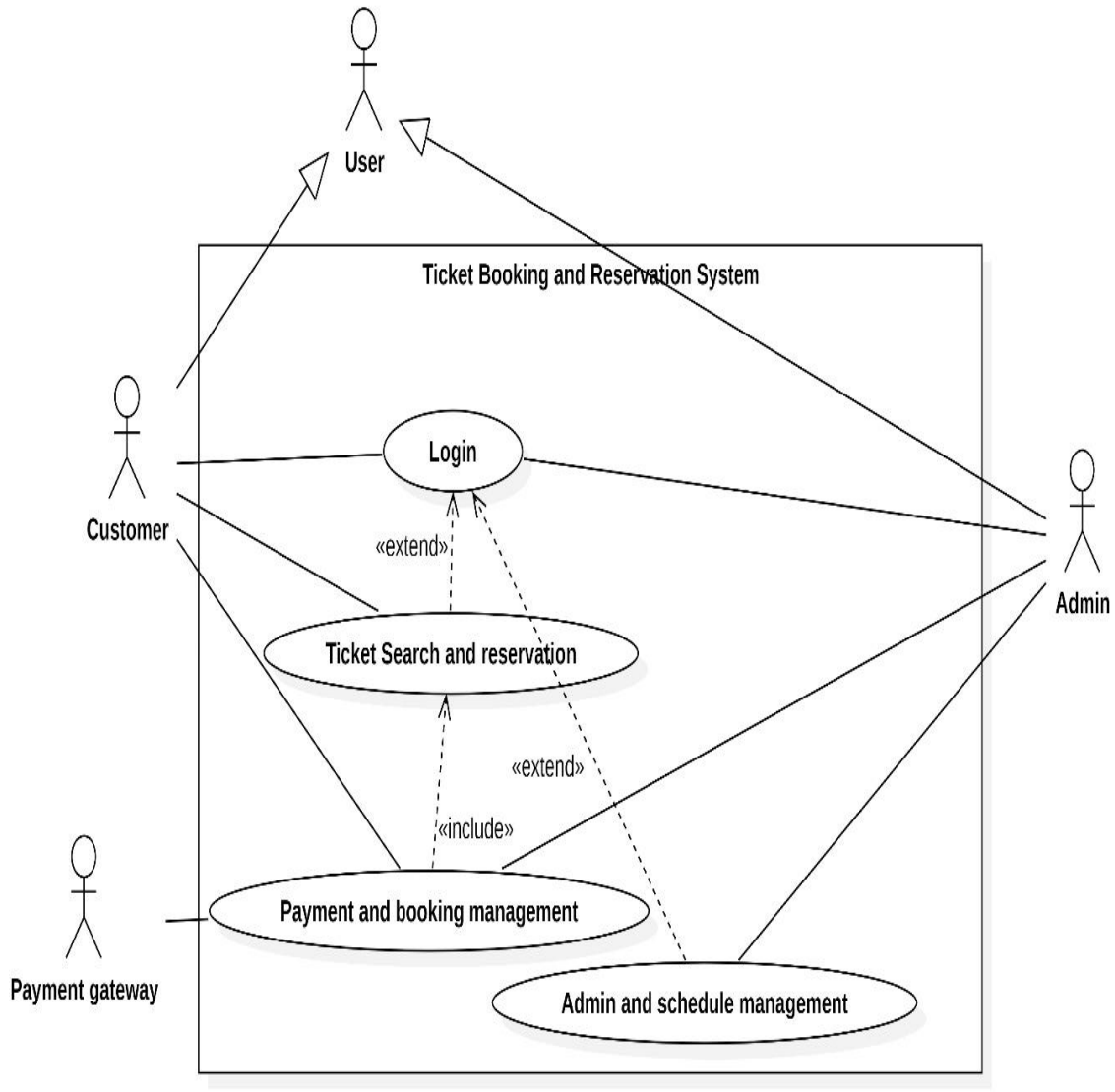
Challenges:

- Updating schedules without affecting existing bookings.
- Managing large volumes of booking data.
- Maintaining data consistency.
- Providing secure administrative access.

2. **Overall Use Case Diagram:**

- Develop an overall use case diagram for the entire application.(Min 4 key usecases)
- Identify and include primary actors (direct users of the system) and secondary actors (external systems or indirect participants). • Ensure the diagram reflects the interaction flow between actors and modules.

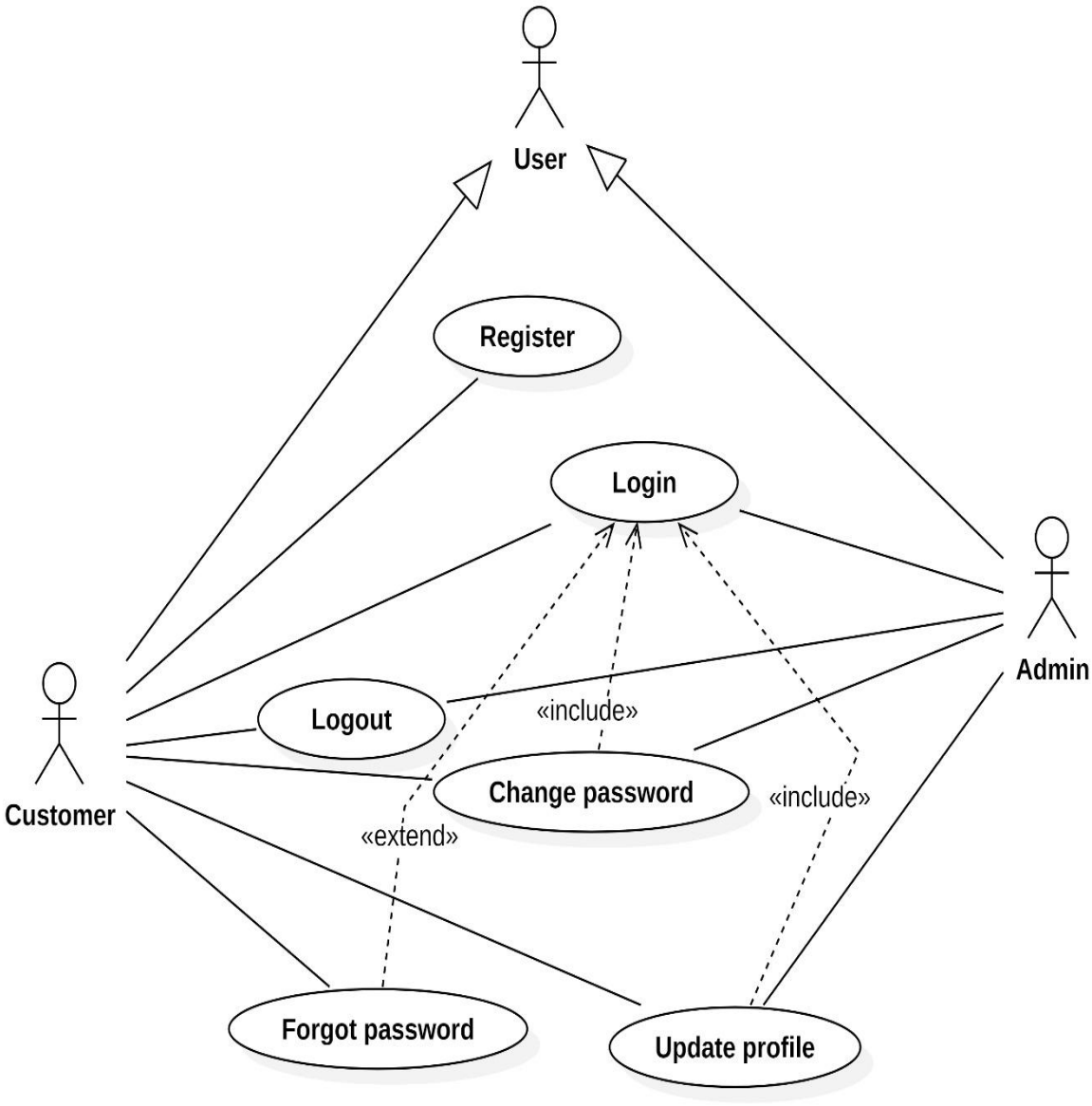
Overall Use Case Diagram



3. Individual Use Cases
   - Create detailed use cases for each module.
   - Apply modelling techniques such as:

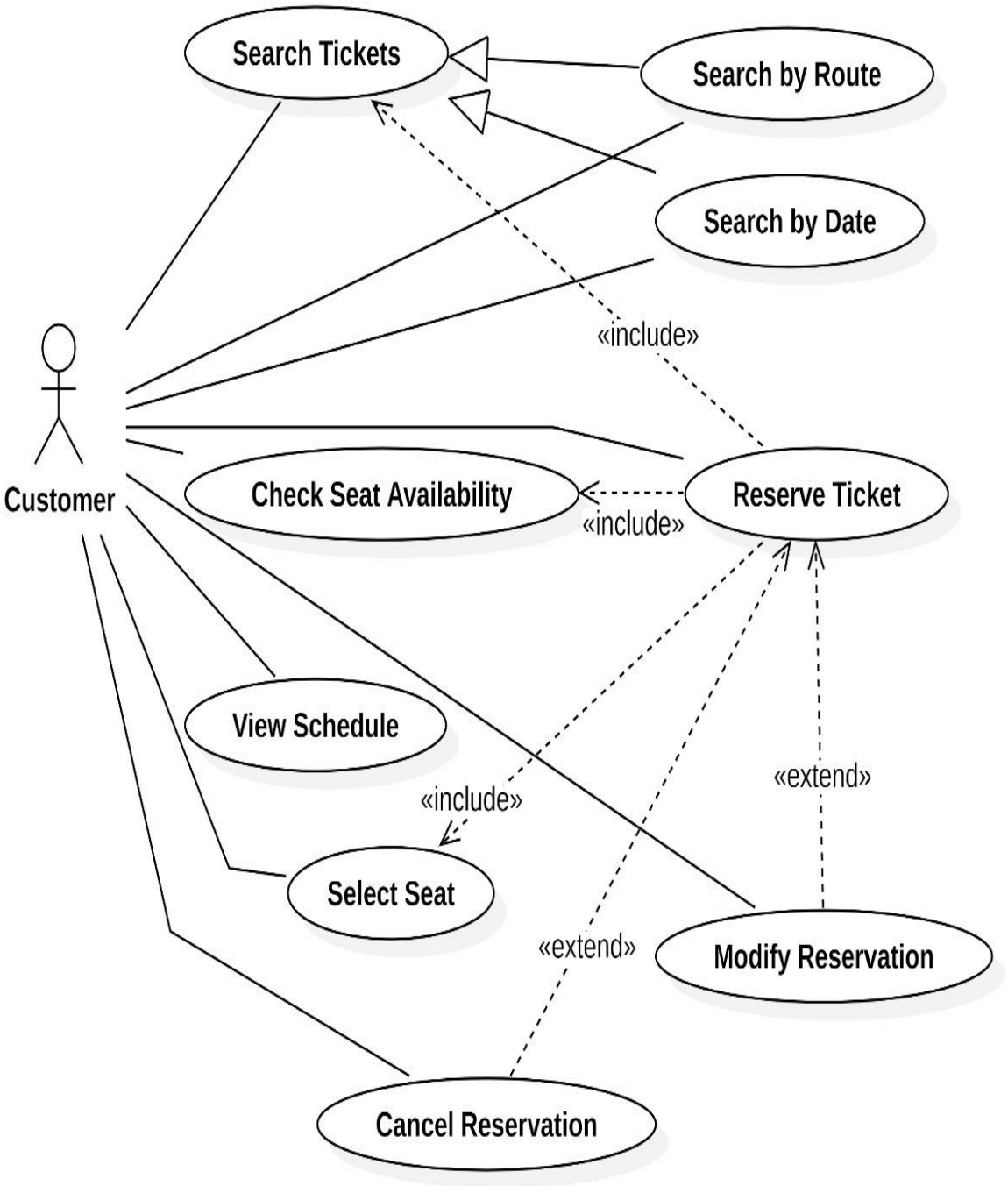
   ♣ Generalization – to represent common behaviors across use cases.
   ♣ Association – to show relationships between actors and use cases.
   ♣ Include – to capture mandatory sub-processes within a use case.
   ♣ Extend – to represent optional or conditional behaviors.

Module 1 – Login & User Management:



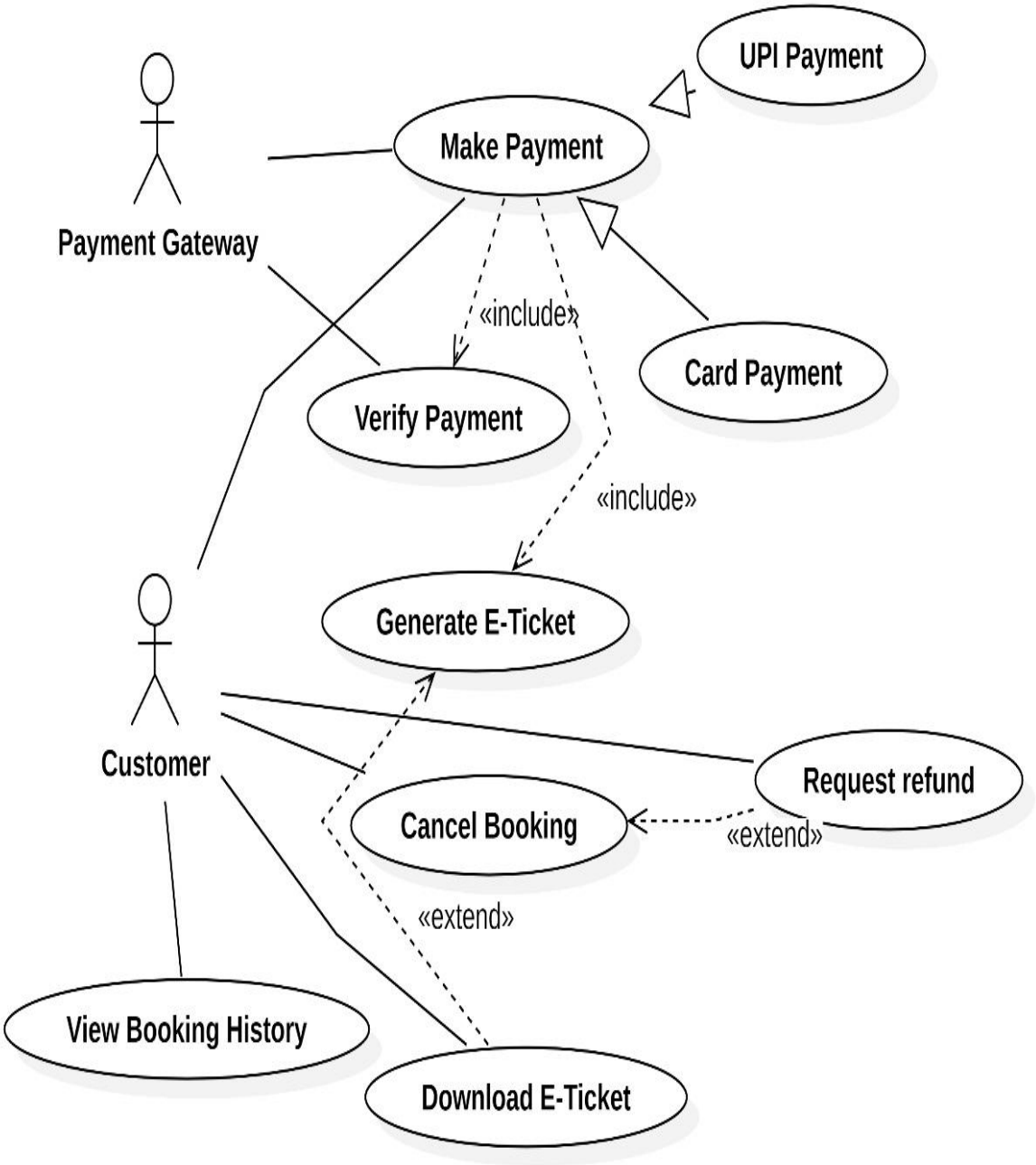
Module 2 – Ticket Search & Reservation

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Module 3 – Payment & Booking Management

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Module 4 – Admin & Schedule Management

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